

An empirical study of content-based recommendation systems in mobile app markets

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ABSTRACT

Recommendation systems are widely used to promote product visibility and sales. However, past research suggests that they primarily benefit market superstars and therefore, can be detrimental to niche products. This is especially critical in superstar dominated markets such as the market of mobile apps. This study incorporates social network analysis and econometric models to empirically examine the impact of content-based filtering (CBF) recommendation systems on the distribution of demand in mobile app markets. The analysis results of two comprehensive panel datasets from App Store and Google Play suggest that CBF recommendations favor niche items and effectively boost the long tail of the market. Moreover, consistent with signaling theory, relative quality signals provided by recommendation systems can impact consumer decision making process, leading to a spillover of demand. Our findings provide important implications for developers and market operators to better promote their products in the highly competitive market of mobile apps.

1. Introduction

The mobile app market is growing rapidly with new app downloads reaching a record high of 230 billion and consumers spending exceeding \$170 billion in 2021 [1]. Apple and Google both have provided their app development and distribution platforms to lower the barriers for entry and create a hypercompetitive market where app visibility and discoverability are the most challenging parts of the mobile app business [2,3]. The large number and the diversity of mobile apps in this market and the constant evolution of the platforms have made it challenging for users to find their preferred apps, for developers to strategically position their apps and differentiate them from competitors, and for market administrators to design effective mechanisms to promote their offerings [4].

Studies show that the mobile app market is dominated by superstars where consumers generally go after apps from well-known developers or the ones that have successfully established their reputation in the market [4], leading to a self-reinforcing loop in which the strong gets stronger and the weak gets weaker [5]. Small and new app developers often find it challenging to gain visibility and compete against these dominating players.

Like most e-commerce platforms, mobile app stores incorporate

recommendation systems to improve the visibility of their offerings. The most common approaches for ranking the unseen items are content-based filtering (CBF) and collaborative filtering (CF). CBF recommendations are generated based on common product attributes and suggest similar items, while CF recommendations are based on past user search (co-view) or purchase (co-purchase) behavior [6]. Studies on CF recommendation systems suggest that these algorithms may increase the overall demand for both popular and niche products [7]. Yet, demand gain for niche items is reportedly smaller than the gain for popular ones. Therefore, implementing a CF recommendation system can be detrimental to the overall market share of niche products [8]. This is especially critical in the superstar dominated market of mobile apps where 1% of products account for 70% of the total market demand [5].

CBF algorithms operate on product similarity or item-to-item correlation [9] and generate substitutive recommendations for which the demand effect is largely understudied [8]. Since products appearing on a CBF recommendation list are often the direct rivals of the main product originally browsed, developers may adjust their product features, improve their position on these lists, and take advantage of the recommendation system design elements to benefit from their presence on recommendation lists. While literature acknowledges the importance of

Abbreviations: CBF, Content Based Filtering; CF, Collaborative Filtering; SNA, Social Network Analysis.

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design elements in recommendation systems [6], past research has largely overlooked their impact on users' decision making process. The designers of a recommendation system may choose to display the brief product features of the recommended options such as user rating and price next to the information of the main product. Viewed as a true indicator of past user experience, user rating is widely accepted as a quality signal and has a strong influence on a costumer's attitude towards a given product [10]. When a CBF recommendation readily displays such quality signals, the user comparison of these signals may cause a diffusion of demand spillover, shifting demand from the main product to its recommended alternatives [6,11]. In our study, demand spillover describes the change in user adoption decision caused by the existence of the recommendation list and the salient characteristics of the featured products such as user rating.

The overarching goal of this research is to enhance our understanding of the impact of CBF recommendation systems and quality signals they display on demand distribution and consumers' information processing in the superstar dominated market of mobile apps. As discussed earlier, CBF recommendations are understudied while their impact on winner-take-all markets can be critical. Besides, while acknowledging the importance of recommendation system design elements, prior literature fails to examine how the signals conveyed by these elements may impact consumer behavior. Therefore, our goal in this study is to fill these research gaps by answering the following research questions:

1. How do CBF recommendation systems affect demand distribution in mobile app markets?
2. How do comparative product quality signals provided by a CBF recommendation system affect users' decision-making process?

Our study aims to answer the above questions by empirically analyzing and comparing the CBF recommendation systems adopted by the world's largest mobile app distribution platforms – Apple App Store (hereinafter called App Store) and Google Play. Following the body of research on the demand effect of recommendation systems [7,12,13], we use social network analysis (SNA) to compute the strength of recommendations. We choose individual apps as the unit of analysis and use two five-month panel datasets consisting of a total of 724,742 observations from 11,181 apps to investigate the flow of demand within the recommendation systems of mobile apps. Our empirical findings suggest that even after controlling for the effect of exogenous advertisements and a comprehensive set of app-level variables, CBF recommendation systems significantly and consistently impact the distribution of demand in mobile app markets. Interestingly, contrary to findings regarding CF algorithms (e.g. [8]), our results indicate that the existence of a CBF recommendation system may in fact benefit the niche products.

The rest of this paper is organized as follows. Section 2 reviews the relevant literature. Section 3 presents the research model and hypotheses. Section 4 describes the data and research methodology. Section 5 summarizes the results from the empirical analysis. Section 6 discusses the findings, contributions, and limitations. Finally, Section 7 presents the conclusion of the study.

2. Literature review

Mobile apps are distributed through centralized marketplaces such as App Store and Google Play whose primary function is to provide a fair platform for developers to compete and an easy way for consumers to find the apps that match their preferences and satisfy their needs [14]. However, the volatility of user demand, the high rate at which new apps are introduced, the constant changes to app categories, and the short app lifecycle make it extremely challenging to match supply and demand [4]. To succeed in this market, developers need to make their products visible, and recommendation systems are widely adopted as a visibility mechanism in both App Store and Google Play.

A recommendation system is configured to form a large web of interconnected pages to help consumers find a product that meets their requirements and/or to encourage them to purchase additional goods [13]. Recommendation systems can dynamically update the list of featured products based on user preferences and market trends and have long been adopted by e-commerce businesses to reduce consumer information overload [13,15] and improve the certainty and quality of their decisions [16]. They also benefit market administrators by increasing upselling and cross-selling opportunities [7]. Finally, they benefit suppliers by enhancing the visibility, awareness, and confidence for their products [8]. The two most common types of recommendation systems are CBF (that provides recommendations based on the characteristics of the focal item), and CF (that suggests items based on the choices of other users with similar profiles). Also, based on the type of suggested products, they can be classified as complementary or substitutive recommendation systems [11].

Popularized by the wide adoption of the Internet, the long tail effect describes an economic trend in which the aggregate sales of niche items equals or exceeds the sales of superstars [7]. Recommendation systems are considered an important tool for fattening the long tail since they match user preferences and facilitate product discovery [7]. While some studies confirm this view [17], others report that the presence of recommendations can reduce sales diversity, making recommendation systems more beneficial to already popular products [7,8,18]. Through analysis and simulation, [15] explain the inconsistencies and argue that while recommendation systems bring diversity to the choices made by each individual, they concentrate the demand for the overall market to a much smaller segment since they recommend the same items to consumers with similar preferences. CF algorithms suffer from popularity bias as they present a list of items viewed by past users and therefore create an "echo chamber" in which the in-demand items continuously become more popular [8]. They also fail to recommend items with limited historical information [15].

Apart from linking product pages, recommendation systems can also display various quality signals alongside each featured product. The excessive volume of available information in mobile app markets can be overwhelming for users and the easy-to-compare quality signals provided by recommendation systems can help resolve this information overload and help users make better decisions [10]. According to signaling theory, in each transaction, the seller party (signaler) may try to send a message (signal) to communicate to the buyer party (receiver) the quality of their product to influence buyer's purchase decision [19]. In the market of mobile apps, the centralized marketplace (such as App Store and Google Play) represents the seller party, and user rating is an indicator of quality signal [10]. User rating or buyer's feedback is a "calculation-based trust" that can lower costumer's risk when the level of familiarity with the product is low [20]. However, signal effectiveness depends on the degree to which a receiver scans the environment for signals and it can be measured by observing the change in receiver behavior [21]. Recommendation systems can conveniently place quality signals of recommended items alongside those of the focal product. Past research suggests that the presence of visible links between products can cause exposure and word-of-mouth spillover which can in turn impact sales [6,13,22,23].

The literature acknowledges that the type of recommendation system is important in determining its demand effect [8]. While CF recommendations are studied extensively, CBF type algorithms are largely understudied. However, CBF architecture and the fact that it operates based on product-to-product correlation [24] means that they can potentially be more beneficial to the sales of niche products. Besides, implicit design choices that go into creating recommendation systems and the signals they convey can have a significant impact on user consumption pattern [8]. But these factors have also not been empirically tested in the literature. Understanding the impact of CBF type algorithm as well as recommendation system signals can be crucial especially in a superstar dominated market such as the mobile apps market.

Considering that the output of recommendations is highly dependent on design decisions and the unique characteristics of the mobile app ecosystem, we use Social Network Analysis (SNA) to study a distinct context – mobile app market's CBF recommendations. Building on [13] incoming/outgoing network approach, we present a comprehensive view of the impact of CBF recommendation systems on the dynamics of consumer choice and demand within the mobile app market. Our work is unique in its context and the use of SNA in examining the competition between the focal app and its neighbors in a recommendation network.

3. Research model and hypotheses

The study of CBF recommendation systems remains scarce in the literature. Past research has focused on CF recommendation systems and concluded that CF recommendation algorithms can only boost market superstars. However, since CBF recommendations establish links between products based on similar attributes (product correlation), instead of putting spotlight on popular products, they are essentially recommending items with consumer-preferred attributes [24]. Therefore, contrary to CF algorithms, they should be capable of providing visibility to the long tail. Moreover, the design choice made by the creators of recommendation systems to display product quality signals (such as user rating scores) along with each recommendation can have a profound impact on consumer's assessment of the existing alternatives and their ultimate decision [8]. However, past research has not empirically examined the effect of recommendation system signals on user behavior.

CBF recommendation systems display featured products related to the one that the user has originally browsed. More appearance on the recommendation list can increase a product's visibility and the probability of sales. However, the strength of recommendation is not only captured by the number of appearances, rather the prominence of the recommender nodes must also be considered. A recommendation from a popular item may help to put the recommended item in a more centralized location in the network and gives it a strategic advantage that other items with comparable number of recommendations from less popular items may not enjoy. This consideration is especially crucial in superstar dominated markets.

Prior studies that focused on CF recommendation algorithms report a positive relationship between the strength of recommendations and demand or sales of products [7,25], and consistent with their findings, we expect a similar relationship in the CBF context and argue that, all else being equal, there is a positive relationship between the strength of recommendation and demand for the focal app:

Hypothesis 1a. The strength of recommendation for a focal app is positively associated with its demand.

Building on H1a, it is also interesting to understand how the CBF recommendation systems impact demand distribution in the market, especially within digital markets where demand distribution is highly skewed towards the superstars. Unlike CF algorithms in which mass consumer behavior can drive overall market demand towards a specific set of items [15], a CBF system is largely driven by product attributes. In CBF algorithms, the recommendations are determined based on the correlation between product attributes. For example, a new and unpopular communication app may offer attributes similar to a popular app such as WhatsApp (e.g., disappearing messages, and end-to-end encryption). Therefore, while it is unlikely for such a relatively unpopular communication app with limited history to appear on CF recommendations, it may gain more visibility through appearing on the CBF recommendation lists.

We argue that outside the closed circle of popular apps that constantly appear on CF recommendation lists, there can be niche products with attributes that better match user preferences and CBF algorithms can ensure the discovery of such relevant items. Therefore, we expect such system to benefit niche products more than superstar (popular) products. We examine the interaction effect of app popularity

(i.e., superstar vs. niche apps) on the relationship between the strength of recommendation and demand for the focal app. We hypothesize:

Hypothesis 1b. App popularity moderates the positive relationship between the strength of recommendation and demand, such that the effect is weaker for superstar apps.

Beyond just the existence of links between product pages, the quality of connections also carries significance. Recommendation systems rank items in a list to guide user's decision-making process. Similar to how occupying a more central location in a physical store or an optimal position on the shelf may result in higher visibility and demand, the position of an item on a recommendation list is an indicator of link quality. Past research suggests that users observe the sorting of items on a recommendation list and their decision could be influenced by the order of the list [26]. When a recommended app occupies a higher position on the list, it is conveniently visible due to its placement (particularly on mobile devices that have a relatively small viewing area), and it garners more attention because of consumers' perception of its relevance [27].

In a CBF recommendation, this ranking shows the relevance of the recommended item and is determined by the probability of user click behavior given a set of user, contextual, and impression features [28]. Thus, we expect that:

Hypothesis 2. The recommendation relevance of the focal app is positively associated with its demand.

Apart from the network related attributes, past research highlights the importance of considering the effect of recommendation system design elements on demand [8], especially in substitutive recommendations where featured items can be the direct competitors of the focal product. While some recommendation systems display the characteristics of featured products (such as user rating and price), others only show a list of featured items with no quality or price cues. Based on the premises of signaling theory, the availability of these product cues can reduce information asymmetry and allow users to quickly assess the focal product against the recommended ones and change their attitude and their subsequent decision in favor of a recommended alternative [29].

In e-commerce, user rating is a proxy for product quality [30]. It describes the quantified opinion of past users and can trigger a herd mentality by impacting the decision of future adopters, especially those who lack sufficient knowledge or experience of the product [31]. In an online marketplace, the presence of such observable quality signals can inform users of existing alternatives and help them choose the best product [10]. When quality signal (i.e., user rating) is readily available, and the focal item has lower quality relative to those of the recommended alternatives, the user may observe the comparable quality signals. Then, this consideration may shift demand from the focal product to the recommended alternatives, causing a diffusion of demand spillover. We expect:

Hypothesis 3. The relative rating of the recommended apps to that of the focal app is negatively associated with the demand for the focal app.

Apart from quality signals, accessible environmental cues about the product such as price can also influence consumer's perception of risk [10]. Mobile apps are experience goods for which the consumer assessment of value takes place after consumption [32]. As a result, users will only know if a certain app fits their preferences after purchasing (downloading) it. The risk of a purchase decision is determined by the monetary cost and the buyer's subjective assessment of an unfavorable outcome.

Consequently, price could impact the way users digest quality signals. When a purchase decision involves high monetary risk, consumers have strong motivation to search for risk reducing informational cues such as quality signals which would reinforce the effect of quality signal consideration on demand. Therefore, relative rating score might have a

greater effect on download decision when the focal app has a high initial price tag. In contrast, users may perceive a low risk of monetary loss when the focal app is free or low priced. This would offset the demand effect of quality signals and demotivate users from paying attention to the rating scores displayed by the recommendation system. We theorize that:

Hypothesis 4. Price moderates the negative relationship between relative rating and focal app demand, such that the effect is stronger for more expensive apps.

Apart from the recommendation network attributes, prior literature has identified a set of app-level factors that impact demand in both Apple and Google's mobile app markets [2,33,34]. Following these works, we control for the effects of price, app age, version age, update frequency, file size, app category, age rating and the existence of in-app purchase options on demand. Moreover, our unique dataset allows us to consider the important and unexplored effect of advertisement on mobile app demand. Fig. 1 shows the proposed conceptual research framework.

4. Methodology

Both Apple and Google incorporate recommendation systems to create visibility (impression) for the apps they offer in their markets. In this study, we focus on CBF recommendation networks, which are displayed on every app's product page regardless of the device used to access the platform [35]. Our outcome variable of interest is the daily demand for an individual app (focal app). First, we examine the impact of CBF recommendations on demand distribution (H1a, H1b, and H2) in both App Store and Google Play. Then, since App Store does not readily provide quality signals on their recommendation system, we focus on Google Play dataset to examine the demand spillover caused by explicit quality signals (H3 and H4). Fig. 2 shows the structure of the network on the two platforms.

4.1. Data

To test our research model, we collected data from App Store and Google Play between Feb 1, 2019, and June 30, 2019 (5 months). Each app on the platforms has an associated product page, which provides detailed information such as app title, description, version, category, rating score, rating count, size, and developer name. Besides, both platforms provide user-specific recommendations under titles such as "Recommended for You" which can be based on user's past search or install behavior (CF recommendation systems) as well as general CBF recommendations that are identical for users in the same geographical location¹ even when the user has not logged in and they are determined based on similarity with the main (focal) app that is being viewed and provide an alternative way for app discovery on the platform. These lists are labeled "You May Also Like" on App Store and "Similar" on Google Play² and are displayed regardless of the device used to access the product page. We collected data from the listing pages of App Store and Google Play. Both platforms publish a daily list of apps for each app category. We based our data collection effort on App Store's category list, and then manually matched the corresponding categories on Google Play. Our web scraper would collect the data for all the apps in the United States within the 24 categories on App Store and their equivalent categories on Google Play.³ Given that the app categories differ slightly

on the two platforms, we combined similar categories to make them comparable across platforms.

Next, in order to control for the effect of external marketing, we collected the detailed information of the advertisement campaigns for each app, including the number, time, and duration of campaigns as well as the selected ad networks from www.sensortower.com, and supplement the information regarding the release date and version history of each app from another source: www.apprtrace.com.

4.2. The proxy for demand

As described earlier in the methodology section, we are interested in examining the various characteristics of the recommendation network on app demand which we operationalize as the number of downloads. However, download count is not readily available in our datasets since App Store and Google Play like many other e-commerce platforms such as Amazon do not reveal their exact demand figures. The rating count which represents the number of people who have downloaded and reviewed the app is recognized in the mobile app literature as the lower bound for app demand [36]. Google Play posts the exact daily rating count figure for each app and therefore, we use this reported value for our Google dataset. However, Apple publishes the exact counts for values <1000 and then rounds the number to the closest 100 for the larger rating counts (e.g., 12.5 K) and keeps this convention up to 1 million. After that, they round the numbers to the closest 100,000 (e.g., 1.1 M). This makes it impossible to know the accurate amount of change in the rating counts for apps that have >1000 ratings. To tackle this problem for apps with over 1000 ratings on the Apple dataset, we operationalize rating count of app i on the day t as a moving average⁴ of the rating count of i from day t to $t + 6$. This allows us to capture the changes in a weekly time window and obtain more accurate approximation for the dependent variable of our Apple dataset. On both datasets, we define app rating count as a proxy for demand and use the daily change of rating count as the target variable of our study.

4.3. Empirical models

For each individual app i on day t , we operationalize all the relevant variables. Dependent variable of our study is $(LOG_DELTA_RCN_{i,t+1})$ which shows the demand for i on day $t + 1$. For testing H1a, we use PageRank to measure the strength of recommendations. PageRank captures the relative importance of nodes in a directed network and operates based on the idea that important nodes have many incoming links from other important nodes [37]. The PageRank value of a node after k steps is the probability of a random walker arriving at that node after k steps. It can be computed using eq. (1):

$$PAGE_RANK_{i,t} = \frac{(1-d)}{n} + d \sum_{j \in G_{(i)}} \frac{PageRank_{j,t}}{Outdegree_{j,t}} \quad (1)$$

In the above equation, $j \in G_{(i)}$ if there is a link from app j to app i on day t (app i appears on the recommendation network of app j) and $Outdegree_{(j,t)}$ is the total number of recommended apps on app j 's product page (outgoing links). d is a damping parameter between 0 and 1 and it is set to 0.9 in this study.

To account for the interaction effect of app popularity (H1b), we compute the median of rating count for the two datasets ($\bar{x}_{Apple} = 4700$, and $\bar{x}_{Google} = 36724.5$) and create the dummy variable $POPULARITY_{i,t}$

¹ The data for this study was collected in the United States.

² App Store displays a maximum of 15 similar apps, while the length of the list on Google Play can vary. To compare the two platforms, we limit our data collection on Google Play to the first 15 apps on the recommendation list.

³ The popularity distributions for the two datasets are presented in Appendix B.

⁴ We also computed the moving median values and compared the results to the moving averages. While the results of the two approaches are largely consistent, the moving median approach generates a larger number of missing values and reduces the sample size by 20% because the recorded changes are stepwise and the number of downloads may remain zero for several consecutive days. Therefore, we choose to report the findings based on moving averages to minimize the number of missing values.

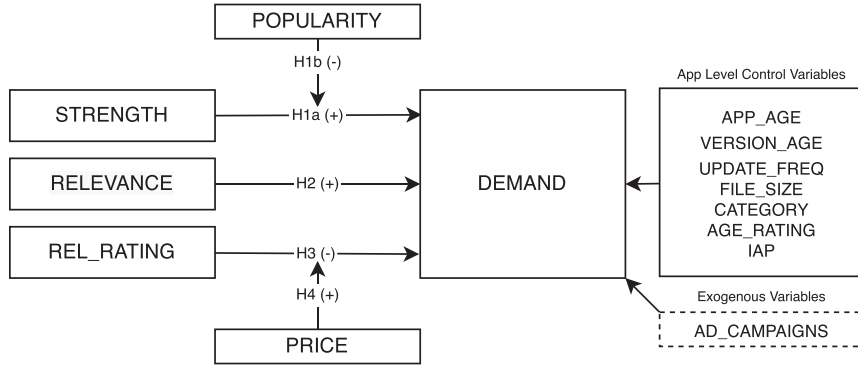


Fig. 1. Research framework.



Fig. 2. Network structure (left), the recommended lists on the two platforms (right).

based on these values.

Apps are shortlisted and ranked based on their relevance score [28], and this score represents the probability of a user clicking on a recommendation link given a set of features such as similarity of keywords, user ratings, app category, total and daily downloads [38].⁵ App i can appear on the recommendation list of other apps such as j and the system assigns a relevance score based on the similarity of i to j . A higher relevance score means higher placement on the list and more visibility for i . However, the exact relevance score is not publicly available and is only manifested by the position of items on the list [28]. Therefore, we use a measure based on Discounted Cumulative Gain (DCG) which is used for measuring the quality of search engine results. DCG assigns a higher value to the items on top of the list and reduces the score at lower ranks using a discount function [39]. We define RELEVANCE value for app i using eq. (2):

$$RELEVANCE_{i,t} = \frac{1}{n} \sum \frac{2^{inv_p} - 1}{\log_2(p+1)} \quad (2)$$

In the above equation, $p \in \{1,2,3,...,15\}$ is the recommendation position of app i ; inv_p is the inverse position of p , and n is the total appearances of p on the recommendation lists.

⁵ The output of the CBF systems (i.e., as symbolized by the “You May Also Like” title on App Store and “Similar” title on Google Play) is not user or device dependent and the list will look identical for all the users in one geographical location (same country IP address).

The two recommendation systems we examine here are different in terms of available design elements. While Google displays user rating and price, App Store only displays a list of recommendations with no quality or price signals. As a result, we test H3 and H4 only on Google dataset. We need to compute a relative measure of quality based on focal app's rating compared to the rating of apps that appear on its recommendation list. To achieve that, we divide the average rating of the recommended apps by the rating of the focal app as shown in eq. (3):

$$REL_RATING_{i,t} = \frac{\sum_{j=1}^n rating_{j,t}}{rating_{i,t} \times n} \quad (3)$$

Where there is a link from app i to app j (app j appears on the recommendation network of app i) and n is the total number of recommended apps on app i 's product page.

Next, $PRICE_{i,t}$ captures the price of app i in US dollars and $AD_CAMPAIGNS_{i,t}$ is the number of advertisement networks on which app i was advertised on day t . Finally, we operationalize the control variables as follows: $RATING_{i,t}$ shows user rating of app i on day t which can be a value between 1 and 5. $APP_AGE_{i,t}$ indicates the number of days since app i was released. $VERSION_AGE_{i,t}$ shows the number of days since the latest update of app i . $UPDATE_FREQ_{i,t}$ captures the frequency of updates for app i and is computed by dividing the number of updates by $APP_AGE_{i,t}$. CAT_CODE_i is a categorical variable that indicates i 's category. $FILE_SIZE_{i,t}$ shows the file size of app i on day t in megabytes. AGE_RATING_i is a categorical variable that indicates the age group for whom the app is suitable. Everyone, Teen, and Mature ratings are

recorded as 0, 1, and 2, respectively. Lastly, IAP_i is a dummy variable that shows whether i has in-app purchase options. We construct three econometric models as shown below and use the full model to test our hypotheses.⁶

$$\bullet \text{ LOG_DELTA_RCNT} = \beta_0 + \beta_1 \text{APP_AGE} + \beta_2 \text{VERSION_AGE} + \beta_3 \text{UPDATE_FREQ} + \beta_4 \text{FILE_SIZE} + \beta_5 \text{CAT_CODE} + \beta_6 \text{AGE_RATING} + \beta_7 \text{IAP} + \beta_8 \text{PRICE} + \beta_9 \text{RATING} + \varepsilon$$

(Control Model)

$$\bullet \text{ LOG_DELTA_RCNT} = \beta_0 + \beta_1 \text{APP_AGE} + \beta_2 \text{VERSION_AGE} + \beta_3 \text{UPDATE_FREQ} + \beta_4 \text{FILE_SIZE} + \beta_5 \text{CAT_CODE} + \beta_6 \text{AGE_RATING} + \beta_7 \text{IAP} + \beta_8 \text{AD_CAMPAIGN} + \beta_9 \text{POPULARITY} + \beta_{10} \text{PAGERANK} + \beta_{11} \text{RELEVANCE} + \beta_{12} \text{REL_RATING} + \beta_{13} \text{PRICE} + \varepsilon$$

(Main Model)

$$\bullet \text{ LOG_DELTA_RCNT} = \beta_0 + \beta_1 \text{APP_AGE} + \beta_2 \text{VERSION_AGE} + \beta_3 \text{UPDATE_FREQ} + \beta_4 \text{FILE_SIZE} + \beta_5 \text{CAT_CODE} + \beta_6 \text{AGE_RATING} + \beta_7 \text{IAP} + \beta_8 \text{AD_CAMPAIGN} + \beta_9 \text{POPULARITY} + \beta_{10} \text{PAGERANK} + \beta_{11} \text{RELEVANCE} + \beta_{12} \text{REL_RATING} + \beta_{13} (\text{REL_RATING} \times \text{PRICE}) + \beta_{14} (\text{POPULARITY} \times \text{PAGERANK}) + \varepsilon$$

(Full Model)

In the above equations, ε is the error term, β_0 is the intercept and β_1 Through β_{14} are the coefficients of study variables.

Table 1 shows the descriptive statistics of all the variables in this study.

5. Analysis and results

5.1. Results

Since our dataset contains both cross-sectional and longitudinal data, it lends itself to estimators specifically designed for panel data. Our analyses are based on generalized least-squares regressions (GLS) with random effect models. The results for both Apple and Google platforms are shown in Table 2.

First, we estimate a random effect model of the change in the rating counts on all the control variables. As shown in columns (A1) and (G1), app-level attributes including *RATING*, *APP_AGE*, *VERSION_AGE*, *UPDATE_FREQ* and *CAT_CODE* similarly affect demand on both platforms. *PRICE* does not significantly predict demand in App Store and is negatively related to demand in Google Play. Also, While *FILE_SIZE* and *IAP* are negatively related to demand on App Store, they both positively affect demand on Google Play. Besides, while *IAP* is positively related to demand on Google Play's dataset, such effect is negative on App Store data. Columns (A2) and (G2) show the main effect model. As expected, *REL_RATING* negatively affects demand on Google Play dataset. However, the hypothesized interaction effect of *REL_RATING* and *PRICE* is not significant. Also, *RELEVANCE* is not a significant predictor of user demand in either dataset. Besides, *AD_CAMPAIGNS* positively affects demand in both markets. However, the effect of *PAGERANK* is not that straightforward. On the App Store dataset, *PAGERANK* is not significant, while the effect is significant and positive on Google Play dataset. Finally, the results for the interaction effect model are presented in columns (A3) and (G3). Once we include the interacting variables, the marginal effect of *PAGERANK* on demand is not solely captured by its respective coefficient (β_{10}). Rather, we must consider the effect of the interacting variables. More specifically, the main effect of this variables must be computed as the partial derivative of the dependent variable. Therefore, H3 is equivalent to computing eq. (4):

$$H3 : \frac{\partial \text{DELTA_RCNT}_{i,t+1}}{\partial \text{PAGE_RANK}_{i,t}} = \beta_{10} + \beta_9 \text{POPULARITY}_{i,t} \quad (4)$$

⁶ We remove *RATING* from our Google Play analyses as we include *REL_RATING* in our main and full models.

Because the interacting variable is a dummy variable, we should interpret the results of group 1 (High Popularity) to the base level 0 (Low Popularity) which is equivalent to the main effects of *PAGERANK* in columns (A3) and (G3). The inclusion of the interaction term makes both the main and interaction effects of *PAGERANK* on demand significant. More importantly, on both platforms the effect of *PAGERANK* is stronger for the low popularity group which is consistent with our theorized relationship. However, in order to fully understand our results, we need to compute the marginal effect of *PAGERANK* at the two levels of our interacting variable (*POPULARITY*). The marginal effect values shown in Table 3 and the interaction plots and marginal plots in Fig. 3 suggest that in both platforms the effect of *PAGERANK* on demand becomes weaker as we move from niche apps to superstars (the smaller gaps for superstar apps). In the main effect model for the Apple dataset, the effect of *PAGERANK* on demand was not found to be significant. However, after adding the interaction term of popularity in the full model, both the main effect and the interaction effect become significant but in opposite directions. Examining the App Store's marginal effects in Table 3, suggests that *PAGERANK* only increases the demand for apps with low popularity and therefore the effect can only be captured when the interaction term is included in the model.

Table 4 summarizes the results of our hypotheses testing.

5.2. Robustness checks

We validate our results by running multiple robustness checks. First, in the full model, we standardized the interacting variable of *PAGERANK* following the recommendations of [40]. We also performed variance inflation factor (VIF) tests and the results reveal that for the full models on both datasets, the VIF values for the interaction and original variables are well below the acceptable threshold values of 20 and 10, respectively [41]. Therefore, we conclude that our models do not suffer from the problem of multicollinearity. VIFs are shown in Appendix A.

Second, the dependent variable of our analysis is $\text{LOG_DELTA_RCNT}_{t+1}$ which is measured at one day after all the independent variables. However, the effect of some variables may take less or more time to realize, and the platforms might have delays in updating the number of rating counts. Therefore, we rerun the model with no-lag and seven-day lag effects. As shown in columns (b) and (c) of Table 5, all the parameter estimates have similar magnitude in the same directions, with similar levels of significance. Hence, the lag effect does not impact our analysis results. Third, to account for the existence of potential serial correlation, we also ran a random effect model with a first-order autoregressive (AR1) disturbance structure. As indicated in column (b) of Table 6, the results are also largely consistent with our main findings. Finally, we standardized all the coefficients to better interpret and compare our results. The results from the standardized models are reported in column (c) of Table 6.

Moreover, to avoid estimation bias due to potential endogeneity that may exist between the interaction term *POPULARITY* and the dependent variable *LOG_DELTA_RCNT*, we collected the download estimates (*DL_EST*) for each app for the month of July 2019 from a third-party website (www.sensortower.com). We only focused on the subset of common apps on both platforms and used the download estimate of the Android app as instrumental variable to capture its popularity on Apple dataset and conversely, used iOS app estimate as instrumental variable in our analysis on Google dataset. We conducted the standard Durbin–Wu–Hausman test using these cross reference instrumental variables. The results of these two tests are shown in Table 7.

Since both test statistics are significant at $p < 0.001$ level, and the F-statistics far exceed the cut-off value of 10 [42], we conclude that *DL_EST* is a valid instrumental variable and control for the potential endogeneity problem using the Control Function (CF) approach [43]. As shown in Table 8, the results of the CF corrected models are consistent with our original models and since endogeneity does not significantly impact our findings, we report and interpret our findings based on our

Table 1
Descriptive statistics of all variables.

Variable	App Store					Google Play				
	Obs.	Mean	Std. Dev.	Min	Max	Obs.	Mean	Std. Dev.	Min	Max
LOG_DELTA_RCNT	307,468	1.532	0.808	0.00	5.00	176,189	1.792	0.98	0.00	6.58
PRICE	462,103	0.094	1.013	0.00	49.99	204,669	0.259	2.51	0.00	99.99
RATING	520,072	4.342	0.593	0.00	5.00	204,669	4.299	0.36	1.00	5.00
APP_AGE	515,697	2171.225	897.394	2.00	4004.00	195,632	1614.965	943.98	9.00	3480.00
VERSION_AGE	514,536	137.050	251.557	0.00	1878.00	199,134	71.406	172.16	−1.00	2919.00
UPDATE_FREQ	514,536	0.029	0.030	0.00	4.00	159,783	0.017	0.03	0.00	0.47
FILE_SIZE	520,072	117.805	128.122	1.00	3100.00	202,310	28.454	38.80	0.00	1371.00
CATCODE	520,072	454.095	258.299	29.00	935.00	204,669	453.155	288.22	20.00	935.00
AGE_RATING	520,072	1.561	1.019	1.00	4.00	199,050	0.117	0.34	0.00	2.00
IAP	520,072	0.499	0.500	0.00	1.00	204,101	0.547	0.50	0.00	1.00
AD_CAMPAIGN	520,072	0.906	2.030	0.00	12.00	204,669	0.392	1.14	0.00	10.00
POPULARITY	520,072	0.504	0.500	0.00	1.00	204,669	0.500	0.50	0.00	1.00
PAGERANK	520,072	0.000	0.000	0.00	0.01	202,697	0.00	0.00	0.00	0.00
RELEVANCE	520,072	0.119	0.120	0.01	1.00	204,669	0.17	0.20	0.01	1.00
REL_RATING	–	–	–	–	–	198,774	1.00	0.08	0.75	4.29

Table 2
Model estimation results (LOG_DELTA_RCNT_{t+1}R).

Variable	APP Store			Google Play		
	(A1)	(A2)	(A3)	(G1)	(G2)	(G3)
	Control	Main effect	Full model	Control	Main effect	Full model
PAGERANK × POPULARITY	–	–	−0.01105*** (0.00)	–	–	−0.00577** (0.00)
REL_RATING × PRICE	–	–	–	–	–	−0.00836 (0.01)
POPULARITY	–	–	0.09656*** (0.00)	–	–	0.24986*** (0.01)
PRICE	−0.00075 (0.00)	0.00023 (0.00)	−0.00093 (0.00)	−0.02455*** (0.00)	−0.02312*** (0.00)	−0.06664*** (0.01)
PAGERANK	–	−0.00088 (0.00)	0.00963*** (0.00)	–	0.00526*** (0.00)	0.00947** (0.00)
REL_RATING	–	–	–	–	−0.51692*** (0.04)	−0.04231*** (0.00)
RELEVANCE	–	0.00492 (0.01)	0.000204 (0.01)	–	−0.01015 (0.01)	−0.00939 (0.01)
AD_CAMPAIGN	–	0.14723*** (0.00)	0.13504*** (0.00)	–	0.36141*** (0.01)	0.32561*** (0.01)
RATING	0.57847*** (0.01)	0.57456*** (0.01)	0.55651*** (0.01)	0.57501*** (0.02)	–	–
APP_AGE	0.00012*** (0.00)	0.00015*** (0.00)	0.00010*** (0.00)	0.00009*** (0.00)	0.00008*** (0.00)	0.00002* (0.00)
VERSION_AGE	−0.00036*** (0.00)	−0.00035*** (0.00)	−0.00036*** (0.00)	−0.00078*** (0.00)	−0.00080*** (0.00)	−0.00083*** (0.00)
UPDATE_FREQ	0.48738*** (0.04)	0.42164*** (0.04)	0.45524*** (0.04)	1.69988*** (0.23)	1.55646*** (0.23)	1.72240*** (0.22)
FILE_SIZE	−0.00006* (0.00)	−0.00008*** (0.00)	−0.00006*** (0.00)	0.00071*** (0.00)	0.00043*** (0.00)	0.00053** (0.00)
CAT_CODE	−0.00036*** (0.00)	−0.00025*** (0.00)	−0.00023*** (0.00)	−0.00050*** (0.00)	−0.00047*** (0.00)	−0.00045*** (0.00)
AGE_RATING	0.00658 (0.01)	0.00013 (0.01)	0.00369 (0.00)	0.02257 (0.02)	0.02227 (0.02)	0.04076** (0.01)
IAP	−0.02952*** (0.01)	−0.03582*** (0.01)	−0.03871*** (0.01)	0.07910*** (0.01)	0.07184*** (0.01)	0.07853*** (0.01)
Constant	−1.36542*** (0.04)	−1.57746*** (0.04)	0.96366*** (0.02)	−1.08085*** (0.09)	1.27248*** (0.03)	1.18825*** (0.02)
Number of Obs.	218,450	218,450	218,450	93,595	90,214	90,682
Number of Groups	4723	4723	4723	3662	3633	3661
R square	0.2523	0.2997	0.3607	0.2181	0.2847	0.3977

Note: In columns (A2), (A3), (G2), and (G3), continuous interacting variable (PAGERANK) is centered. REL_RATING and PRICE are centered in (G3). Standard errors in parentheses. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. In columns (A1–G1) RATING is centered to avoid multicollinearity issue.

Table 3
Marginal effects.

PAGERANK Over POPULARITY	App Store		Google Play	
	dy/dx	Confidence intervals	dy/dx	Confidence intervals
		Min Max		Min Max
Low popularity	0.0096** (0.00)	0.0048 0.0144	0.0096*** (0.00)	0.0065 0.0127
High popularity	−0.0014 (0.00)	−0.0029 0.0001	0.0027* (0.00)	0.0003 0.0051

Note: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

original analyses.

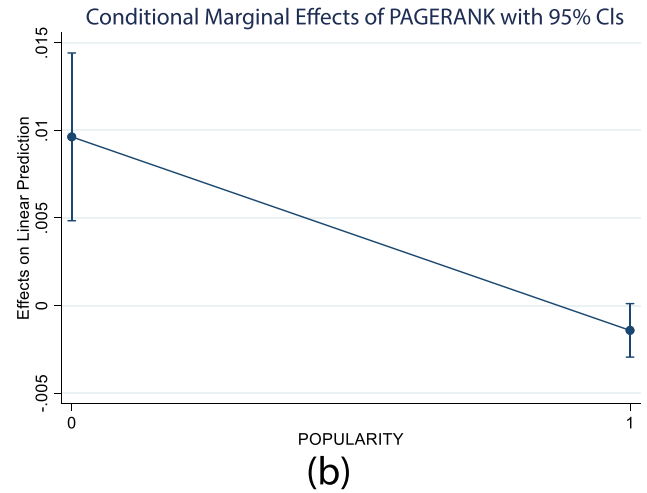
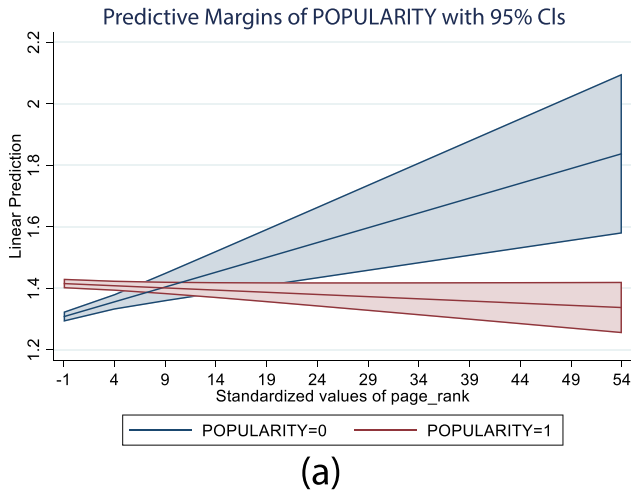
6. Discussion

6.1. Summary of findings

The overarching goal of this study is to enhance our understanding of the impact of CBF algorithms on user demand distribution and decision making in the superstar dominated market of mobile apps. First, we

found that the strength of recommendation positively impacts the focal app's market demand (H1a). Besides, consistent with our theoretical conjecture and the theory of long tail, the presence of CBF recommendation systems alters the demand distribution of the market in favor of less popular products (H1b). However, our conjecture regarding the positive impact of recommendation relevance on demand (H2) was not supported. Mobile app markets can be accessed using phones, tablets, and PCs and the recommended list can be displayed either horizontally or vertically depending on the type of the device and the size of its

App Store



Google Play

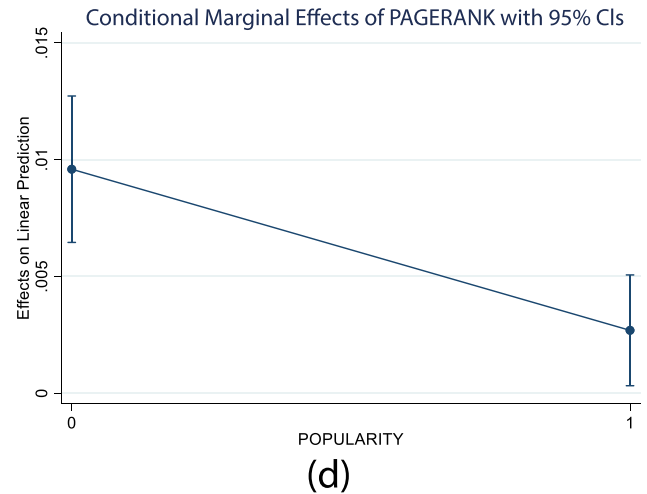
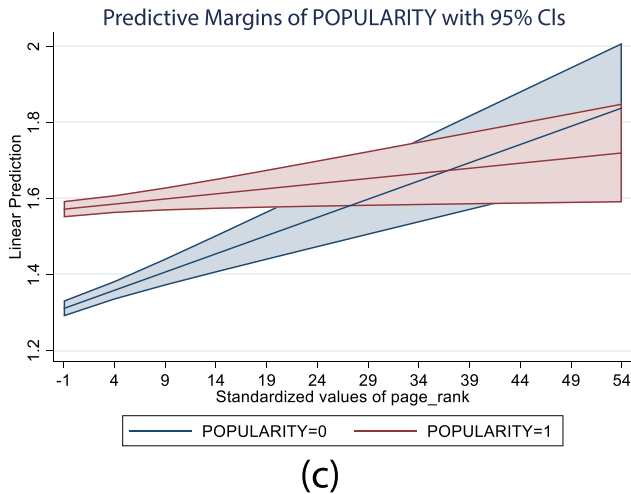


Fig. 3. Interaction and marginal plots.

Table 4
Results of hypotheses testing.

#	Prediction	Outcome	
		App Store	Google Play
H1a	The strength of recommendation for a focal app is positively associated with its demand.	Supported	Supported
H1b	App popularity moderates the positive relationship between strength of recommendation and demand, such that the effect is weaker for superstar apps.	Supported	Supported
H2	The recommendation relevance of the focal app is positively associated with its demand.	Not supported	Not supported
H3	The relative rating of the recommended apps to that of the focal app is negatively associated with the demand for the focal app.	NA	Supported
H4	Price moderates the negative relationship between relative rating and focal app demand, such that the effect is stronger for more expensive apps.	NA	Not supported

screen. We suspect these differences might explain why recommendation relevance was not found significant in either of the two platforms.

Moreover, the effect of relative rating on demand was supported (H3). Consistent with our conjecture, this finding suggests that mobile app users scan the environment for signals and the presence of quality signals on a substitutive recommendation system can persuade them to choose the recommended product with higher quality signal over the one they originally browsed. By facilitating the comparison between the focal app and its relevant alternatives, the recommendation system enables users to confidently make quality decisions [16]. However, regarding the interaction effect of price and relative rating (H4), we did not find support. This could be due to the fact that about 97% of apps on Google Play are free to download [44]. Also, a post hoc examination of the dataset shows that the relative price difference for about 98% of the apps is <2 dollars. Therefore, users might not be sensitive to the price cues they receive from the recommendation system.

Our results also suggest that advertisement is strongly related to increased demand in both markets. Moreover, our findings regarding the impact of app-level factors on demand are largely consistent with previous research. App age is positively related to demand in both markets signifying the importance of app maturity on consumers' decision process [34]. Besides, while the frequency of updates increases demand, the

Table 5
Robustness checks (1).

Variable	(a) Preferred models		(b) No lag		(c) 7-day lag	
	App Store	Google Play	App Store	Google Play	App Store	Google Play
PAGERANK × POPULARITY	−0.01105*** (0.00)	−0.00577** (0.00)	−0.01128*** (0.00)	−0.01408*** (0.00)	−0.01831*** (0.00)	−0.00606* (0.00)
REL_RATING × PRICE	−	−0.00836 (0.01)	−	−0.00132 (0.01)	−	−0.00026 (0.01)
POPULARITY	0.09656*** (0.00)	0.24986*** (0.01)	0.10751*** (0.00)	0.29430*** (0.01)	0.10252*** (0.00)	0.33094*** (0.01)
PRICE	−0.00093 (0.00)	−0.06664*** (0.01)	−0.00196 (0.00)	−0.07344*** (0.01)	−0.00257 (0.00)	−0.070261*** (0.01)
PAGERANK	0.00963*** (0.00)	0.00947** (0.00)	0.00882*** (0.00)	0.00710** (0.00)	0.02203*** (0.00)	0.00249 (0.00)
REL_RATING	−	−0.04231*** (0.00)	−	−0.08504*** (0.00)	−	−0.04856*** (0.00)
RELEVANCE	0.000204 (0.01)	−0.00939 (0.01)	0.00014 (0.01)	−0.00643 (0.01)	0.00037 (0.01)	−0.00742 (0.01)
AD_CAMPAIGN	0.13504*** (0.00)	0.32561*** (0.01)	0.14735*** (0.00)	0.29318*** (0.01)	0.15042*** (0.00)	0.29633*** (0.01)
RATING	0.55651*** (0.01)	−	0.59004*** (0.01)	−	0.37465*** (0.01)	−
APP_AGE	0.00010*** (0.00)	0.00002* (0.00)	0.00021*** (0.00)	0.00016*** (0.00)	0.0002*** (0.00)	0.00007*** (0.00)
VERSION_AGE	−0.00036*** (0.00)	−0.00083*** (0.00)	−0.00033*** (0.00)	−0.00071*** (0.00)	−0.00026*** (0.00)	−0.00039*** (0.00)
UPDATE_FREQ	0.45524*** (0.04)	1.72240*** (0.22)	0.58220*** (0.04)	3.42062*** (0.25)	0.47311*** (0.05)	2.40203*** (0.30)
FILE_SIZE	−0.00006*** (0.00)	0.00053** (0.00)	−0.00004 (0.00)	0.00050** (0.00)	−0.00001 (0.00)	0.00035 (0.00)
CAT_CODE	−0.00023*** (0.00)	−0.00045*** (0.00)	−0.00018*** (0.00)	−0.00057*** (0.00)	−0.00021*** (0.00)	−0.00059*** (0.00)
AGE_RATING	0.00369 (0.00)	0.04076** (0.01)	0.00651 (0.00)	0.06933*** (0.02)	0.02696*** (0.00)	0.09909*** (0.02)
IAP	−0.03871*** (0.01)	0.07853*** (0.01)	−0.03523*** (0.01)	0.06819*** (0.01)	−0.01076 (0.01)	0.08132*** (0.02)
Constant	0.96366*** (0.02)	1.18825*** (0.02)	0.69675*** (0.02)	2.35756*** (0.05)	0.68516*** (0.02)	2.02188*** (0.06)
Number of Obs.	218,450	90,682	270,798	135,287	210,635	77,701
No. of Groups	4723	3661	4924	4153	4674	3039
R square	0.3607	0.3977	0.3343	0.3767	0.2908	0.4229

Note: Standard errors in parentheses. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Table 6
Robustness checks (2).

Variable	(a) Preferred models		(b) Serial correlation		(c) Standardized	
	App Store	Google Play	App Store	Google Play	App Store	Google Play
PAGERANK × POPULARITY	−0.01105*** (0.00)	−0.00577** (0.00)	−0.00467*** (0.00)	−0.00432* (0.00)	−0.01368*** (0.00)	−0.00706** (0.00)
REL_RATING × PRICE	−	−0.00836 (0.01)	−	−0.01977 (0.01)	−	−0.00836 (0.01)
POPULARITY	0.09656*** (0.00)	0.24986*** (0.01)	0.00584*** (0.00)	0.47644*** (0.01)	0.11952*** (0.01)	0.26035*** (0.01)
PRICE	−0.00093 (0.00)	−0.06664*** (0.01)	−0.00325 (0.00)	−0.07201*** (0.01)	−0.00116 (0.01)	−0.06664*** (0.01)
PAGERANK	0.00963*** (0.00)	0.00947** (0.00)	52.91*** (12.81)	0.01197*** (0.00)	0.01192*** (0.00)	0.00947** (0.00)
REL_RATING	−	−0.04231*** (0.00)	−	−0.04950*** (0.00)	−	−0.04231*** (0.00)
RELEVANCE	0.000204 (0.01)	−0.00939 (0.01)	0.00326 (0.01)	−0.00636 (0.01)	0.00030 (0.00)	−0.00177 (0.00)
AD_CAMPAIGN	0.13504*** (0.00)	0.32561*** (0.01)	0.12734*** (0.00)	0.28937*** (0.01)	0.33930*** (0.01)	0.37572*** (0.01)
RATING	0.55651*** (0.01)	−	0.50992*** (0.01)	−	0.40887*** (0.00)	−
APP_AGE	0.00010*** (0.00)	0.00002* (0.00)	0.00008*** (0.00)	0.00001 (0.00)	0.11619*** (0.01)	0.01974* (0.01)
VERSION_AGE	−0.00036*** (0.00)	−0.00083*** (0.00)	−0.00036*** (0.00)	−0.00082*** (0.00)	−0.11133*** (0.00)	−0.14603*** (0.01)
UPDATE_FREQ	0.45524*** (0.04)	1.72240*** (0.22)	0.47853*** (0.04)	1.88928*** (0.23)	0.01696*** (0.00)	0.04567*** (0.01)
FILE_SIZE	−0.00006*** (0.00)	0.00053** (0.00)	0.00005* (0.00)	0.00069*** (0.00)	−0.01009** (0.00)	0.02123** (0.01)
CAT_CODE	−0.00023*** (0.00)	−0.00045*** (0.00)	−0.00020*** (0.00)	−0.00045*** (0.00)	−0.07295*** (0.01)	−0.13415*** (0.01)
AGE_RATING	0.00369 (0.00)	0.04076** (0.01)	0.00745 (0.00)	0.06016*** (0.02)	0.00466 (0.01)	0.01413** (0.01)
IAP	−0.03871*** (0.01)	0.07853*** (0.01)	−0.04037*** (0.01)	0.08703*** (0.01)	−0.02396*** (0.00)	0.04002*** (0.01)
Constant	0.96366*** (0.02)	1.18825*** (0.02)	0.96485*** (0.02)	1.12432*** (0.03)	−0.46933*** (0.01)	−0.56347*** (0.01)
Number of Obs.	218,450	90,682	218,450	90,682	218,450	90,682
No. of Groups	4723	3661	4723	3661	4723	3661
R square	0.3607	0.3977	0.3984	0.4886	0.3607	0.3977

Note: Standard errors in parentheses. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Table 7
Endogeneity tests.

Test	Apple	Google
Durbin Score chi2	650.063***	762.756***
Wu-Hausman	663.28***	803.063***
F	31,870	14,898

increase in the number of days since the release of the last update is negatively related to demand on both markets. These results are in line with prior findings that developers may use updates to increase their downloads [45] while the time gap between updates (i.e. version age) can lower the app's chances for success [36]. Concerning file size, while larger apps have higher demands on Google Play as larger files may give users the impression of more features [4], file size and demand have a

negative relationship on App Store. The app category is also an important determinant of demand [4,34]. In our dataset, the 'Games' category has the highest demand on App Store, and the 'Tools' category has the highest demand in Google Play. Moreover, we found that age restrictions are related to demand only on Google Play with 'Teens' rating having the highest number of downloads [4]. Finally, the existence of an in-app purchase option (IAP) which is indicative of the freemium revenue model is positively related to demand in the Google Play dataset and is consistent with previous studies [4,34]. However, the relationship is negative and significant on the Apple dataset.

6.2. Theoretical contributions

This paper contributes to the literature on the use of recommendation systems as decision aids [46]. It shows that the effect of

Table 8
Robustness checks (3).

Variable	(a)		(b)	
	Preferred models		CF corrected models	
	App Store	Google Play	App Store	Google Play
PAGERANK $\times$ POPULARITY	−0.01105*** (0.00)	−0.00577** (0.00)	−0.00771** (0.00)	−0.01701** (0.00)
REL_RATING $\times$ PRICE	−	−0.00836 (0.01)	−	−0.03758 (0.05)
POPULARITY	0.09656*** (0.00)	0.24986*** (0.01)	2.95613*** (0.56)	6.87351*** (0.79)
PRICE	−0.00093 (0.00)	−0.06664*** (0.01)	−0.00796 (0.06)	0.01160* (0.01)
PAGERANK	0.00963*** (0.00)	0.00947** (0.00)	0.00806** (0.00)	0.07726*** (0.01)
REL_RATING	−	−0.04231*** (0.00)	−	−0.01296*** (0.00)
RELEVANCE	0.000204 (0.01)	−0.00939 (0.01)	−0.00204 (0.02)	−0.00327 (0.04)
AD_CAMPAIGN	0.13504*** (0.00)	0.32561*** (0.01)	0.06492*** (0.02)	0.39754*** (0.05)
RATING	0.55651*** (0.01)	−	−0.32562 (0.24)	−
APP_AGE	0.00010*** (0.00)	0.00002* (0.00)	0.00018*** (0.00)	0.00078 (0.00)
VERSION_AGE	−0.00036*** (0.00)	−0.00083*** (0.00)	−0.00066** (0.00)	−0.00107*** (0.00)
UPDATE_FREQ	0.45524*** (0.04)	1.72240*** (0.22)	1.0006*** (0.29)	1.72072*** (0.36)
FILE_SIZE	−0.00006*** (0.00)	0.00053** (0.00)	−0.00151*** (0.00)	0.01067*** (0.00)
CAT_CODE	−0.00023*** (0.00)	−0.00045*** (0.00)	0.00011 (0.00)	−0.00167*** (0.00)
AGE_RATING	0.00369 (0.00)	0.04076** (0.01)	0.06309 (0.02)	−1.04945*** (0.16)
IAP	−0.03871*** (0.01)	0.07853*** (0.01)	−0.34360*** (0.04)	−1.05914*** (0.13)
vR ₂ R.POPULARITY	−	−	−3.00076 (0.56)	−6.58512*** (0.79)
Constant	0.96366*** (0.02)	1.18825*** (0.02)	−0.52364 (0.33)	−0.71333** (0.25)
Number of Obs.	218,450	90,682	31,885	14,913
No. of Groups	4723	3661	567	406
R square	0.3607	0.3977	0.3838	0.4440

Note: Standard errors in parentheses. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

recommendation system on market demand may depend on the type of algorithm. It further examines the role of recommendation system design elements on user decision and demand spillover.

First, we extend the knowledge on the demand effect of recommendation systems by showing that unlike CF algorithms, CBF recommendations can help decrease market inequality by boosting demand for niche items consistent with the long tail theory. Second, we advance the understanding on how the presence of quality signals impacts demand spillover in substitutive recommendations. Mobile app users are unlikely to know the true quality of an app prior to download, and in the absence of explicit product information, users tend to rely on quality signals that they perceive as trustworthy. User rating is considered a reliable, high cost quality signal as platforms actively fight off spammers and fake ratings [20]. Past research shows that user rating cues can trigger herd behavior and influence user adoption decision [30]. Our study is first to examine how users observe and evaluate comparable quality signals provided by a recommendation system. Our findings show that when recommendation systems readily display these signals for substitutable items, users may change their original choice and follow system recommendations, leading to a diffusion of demand spillover. Lastly, our study contributes to the mobile app literature by identifying and validating the network-level predictors of demand as well as showing the influence of advertising efforts.

6.3. Practical implications

From a practical standpoint, our study provides insights for developers and platform providers regarding the critical issue of app discovery. For developers of niche apps, it is important to recognize the value of CBF recommendations and focus on the network status of their apps within each app cluster. They may take measures to increase their presence and improve their position on CBF recommendations by using strategies such as app store optimization (ASO). Moreover, our finding regarding the effect of quality signals suggests that users actively scan for these signals when they are browsing through product pages, and they observe and compare the signals procured by the recommendation system vis-à-vis the focal product to resolve the issue of information overload and make better decisions. Understanding the effect these signals have on user's decision-making process can help mobile app developers optimize their product pages and increase their success in the

race for winning user attention. Our result indicates that the appearance of highly rated apps on recommendation lists can cannibalize demand for a focal app and since Google allows developers to place paid ads on the listing pages of the platform, developers can use this knowledge to strategically place their ads to benefit from users' herd mentality. For platform owners, enhancing visibility algorithms to boost quality niche apps can encourage innovations and increase long term profitability.

6.4. Limitations and future directions

As with any empirical research, this study is not without limitations. The distribution of download and usage in mobile app markets is right skewed. For instance, from 2.8 million available apps on Google Play, 2.3 million have <1000 downloads while the number of apps with over 1 million downloads is <40,000 [47]. In our data collection effort, we included apps from all categories from App Store and a comparative sample from Google Play. However, our dataset still captures a small portion of the entire network and therefore can be subject to the usual caveat of a truncated sample. Besides, the exact list of factors that go into determining the apps that appear in "You May Also Like" on App Store and "Similar" on Google Play are not disclosed by Apple or Google. But app category, description, screenshots, and keywords are some of the factors identified by previous studies [48], which indicate that these recommendations are the result of CBF algorithms. However, geographical location of users may change the recommendation results since marketing strategies or sociopolitical reasons can affect the presence of some apps in certain markets. Therefore, we acknowledge the limitations of our dataset and caution that our findings may only be interpreted for a single market, specifically, the US market where the data were collected. Furthermore, as is the case with most network studies, our estimation model also has potential endogeneity issues. However, we tried to address these concerns by running multiple robustness checks. To better understand consumers' behavior, future research should try to compare the influence of CF and CBF algorithms in the same market. Another possible direction for future research is to examine users' scope of attention to top ranking lists as well as their evaluation of recommendations provided by the platform through survey and experiment methodologies.

7. Conclusions

This study examines the role of CBF recommendations on consumer choice and demand in the mobile app markets. Our analysis of the data collected from two major app markets suggests that despite the prior findings regarding the impact of CF recommendation systems on demand distribution [8], CBF algorithms in the mobile app market are effectively boosting niche apps and provide visibility to the long tail of the market. As one of the first studies that analyze comparable datasets on both Google Play and App Store – the two leading mobile app platforms, our study is timely and novel in examining the impact of CBF recommendation systems on the distribution of demand in the mobile app markets. The fact that the first two hypotheses are consistently supported across platforms demonstrates the robustness of our findings and lends strong support for market operators, developers, and other researchers to develop effective strategies and policies based on our results. Moreover, by examining recommendation system design elements we provide insights into the process of decision making of mobile app users when they are provided with substitutive recommendations. Our findings show that the display of quality signals by a recommendation system can trigger user herd mentality and shift demand from a

focal product to a highly rated one displayed on the recommendation list.

CRediT authorship contribution statement

Mohsen Jozani: Conceptualization, Methodology, Data curation, Formal analysis, Writing - original draft, Writing - review & editing. **Charles Zhechao Liu:** Conceptualization, Methodology, Supervision, Writing – review & editing. **Kim-Kwang Raymond Choo:** Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A

Table A
Variance inflation factors (VIFs).

Variable	Apple dataset		Google dataset	
	VIF	1/VIF	VIF	1/VIF
pagerank	4.06	0.25	2.47	0.40
popularity	1.61	0.62	3.11	0.32
price	1.02	0.98	1.91	0.52
pagerank × popularity	3.81	0.26	2.45	0.40
rel_rating × price	–	–	1.35	0.74
rel_rating	–	–	1.20	0.83
rating	1.40	0.71	–	–
relevance	1.00	1.00	1.73	0.57
ad_campaign	1.27	0.79	1.35	0.74
app_age	1.17	0.85	4.55	0.22
version_age	1.19	0.84	1.26	0.80
up_freq	1.40	0.72	1.59	0.63
size	1.10	0.91	2.59	0.39
catcode	1.10	0.91	2.76	0.36
agerating	1.10	0.91	1.24	0.81
iap	1.11	0.91	2.54	0.39
Mean VIF	1.60		2.14	

Appendix B

We built a comprehensive dataset that entails a wide range of apps with varying popularity. This can be demonstrated through the distribution of rating counts for the two datasets. The rating count represents the number of people who have downloaded and reviewed each app.

While on both datasets, there are apps with only 1 rating, 16% of apps on Apple and 7% of apps on Google dataset have <100 ratings. Going up to 500 ratings, the figures are 30% and 17% on Apple and Google datasets, respectively.

Table B.1 shows the frequency distribution of the popularity of apps in the two datasets.

Table B.1
App popularity frequency distribution.

Dataset	Min	Max	Percentiles								
			1%	5%	10%	25%	50%	75%	90%	95%	99%
Apple	1	11,600,000	11	41	103	667	4700	24,300	103,100	228,200	1,000,000
Google	1	88,400,000	14	106	375	4625	36,725	223,571	967,640	2,267,701	10,600,000

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